

CHETANKUMAR KHADKE

AI & Analytics Translator | Staff Software Engineer

New Jersey, USA | +1 848-348-2810 | khadkechetan@gmail.com

linkedin.com/in/khadke-chetan | khadkechetan.github.io/khadkechetan | medium.com/@khadkechetan

PROFESSIONAL SUMMARY

AI and analytics translator with 13 years connecting business problems to analytics and AI solutions across customer operations, supply chain and trade documentation, and enterprise data discovery. Identifies and prioritizes high-value opportunities, translates them into analytics use cases and product requirements, and bridges business stakeholders with analysts, data engineers, and developers. Conceptually fluent from OLAP and dimensional modeling through logistic regression and classical supervised learning to deep learning and large language models, with the judgment to match technique to business problem and cost. Currently leads AI initiatives at Xpanse for a customer document operations platform spanning 500+ document types and millions of documents weekly, delivering a ~30% accuracy gain and a sharp reduction in manual review.

TECHNICAL SKILLS

Business & Analytics Translation: Business problem framing, opportunity identification and prioritization, analytics use case definition, requirements translation, stakeholder management, solution roadmapping, change management and adoption, value realization and ROI tracking, KPI and success-metric definition, executive communication

Analytics, Data & BI: OLAP and dimensional modeling, star schema design, data warehousing, ETL/ELT, Amazon Redshift, BigQuery, SQL, MySQL, MongoDB, Neo4j, Elasticsearch, Knowledge Graphs, Tableau, Power BI, Looker, dashboard and metric design, cohort and trend analysis, experiment design

Machine Learning: Logistic regression, SVM, Naive Bayes, decision trees, clustering, time-series forecasting, deep learning (CNN, RNN, Transformers), feature engineering, model selection, evaluation, and business fit

Large Language Models (LLM) & Generative AI: Gemini 3 Pro, Gemini Flash, Claude Sonnet on Amazon Bedrock, Llama 3.2, Mistral AI, OpenAI, Qwen 2.5 VL, Hugging Face Transformers, LangChain, LlamaIndex, Fine-Tuning, Prompt Engineering, RAG, Hallucination Mitigation, Guardrails, Model Evaluation

Agentic AI: Multi-Agent Systems, Agent Orchestration, LangGraph, Stateful Agent Graphs, Tool/Function Calling, Model Context Protocol (MCP), Agentic RAG, Human-in-the-Loop, Agent-to-Agent (A2A) Communication, Planning & Reasoning, Memory Management

NLP & Document AI: LayoutLM, Donut, UDOP, BERT, DocVQA, Named Entity Recognition (NER), Named Entity Linking (NEL), Text Classification, Topic Modeling, Sentiment Analysis, Embeddings, Vector Search, Semantic Similarity

MLOps & Cloud: AWS (SageMaker, Bedrock, Bedrock AgentCore, AgentCore Runtime, Strands Agents SDK, ECR, ECS, S3, CloudWatch), GCP, Docker, MLflow, Git, CI/CD, Model Deployment Pipelines, Model Monitoring

Programming Languages: Python, R, Java, SQL

EDUCATION

Bachelor of Technology, Computer Science

Dharmsinh Desai University, Nadiad, Gujarat, India

2009 - 2013

CERTIFICATIONS & AWARDS

- Machine Learning (Silver Medal) and Deep Learning - Indian Institute of Technology Madras (NPTEL)
- Bravo Award, received 3 times for outstanding work - Persistent Systems
- XOR-Champ Award for outstanding work - Xoriant Solutions

RESEARCH PUBLICATIONS

- Khadke C., Rana S., Shah V. "Recovering Workflows from Functional Tests" - IEEE ICPC, Florence, Italy, 2015.
- Shah V., Khadke C., Rana S. "Mining Process Models and Architectural Components from Test Cases" - IEEE ICSTW, Graz, Austria, 2015.
- Shah V., Khadke C., Rana S. "Knowledge Transition: Discovering Models from Functional Tests" - IEEE ISSRE, Gaithersburg, MD, USA, 2015.
- Khadke C. "Knowledge Acquisition (KnowX)" tool demonstration - ENASE, Rome, Italy, 2016.

Full publication list: dblp.uni-trier.de/pers/hd/k/Khadke:Chetan

Technical writing: medium.com/@khadkechetan

PROFESSIONAL EXPERIENCE

Staff Software Engineer

Feb 2025 - Present

Xpanse

New Jersey, USA

Customer document operations platform automating classification and data extraction across 500+ document types and 500+ data fields, processing millions of documents weekly.

- Partner with customer operations and product stakeholders to identify and prioritize high-value automation opportunities, translating manual document-handling pain points into a sequenced analytics and AI roadmap with agreed success metrics.
- Led LLM integration for the production extraction pipeline, raising end-to-end field extraction accuracy by approximately 30% across 500+ field types and materially reducing manual review effort per document.
- Serve as the bridge between business stakeholders, data engineers, and application developers, converting business rules and field-level requirements into technical specifications, acceptance criteria, and model evaluation plans.
- Designed and shipped a production multi-agent orchestration layer on LangGraph, coordinating classification, extraction, and validation agents through tool/function calling, with Model Context Protocol (MCP) integrations exposing platform tools and document context through a standard interface.
- Introduced context-aware guardrails, confidence thresholds, and automated validation checks that sharply reduced hallucination rates and produced auditable, business-defensible outputs on business-critical fields.
- Drove adoption and change management with operations teams, defining human-in-the-loop review workflows and confidence-based routing so staff could trust, verify, and act on model output.
- Built a framework that generates field-specific prompts on demand from document-field mappings, significantly cutting onboarding time for new document types and new customers.

Xoriant Solutions

May 2021 - Feb 2025

Software Engineer

Aug 2024 - Feb 2025 | New Jersey, USA

Supply chain and trade document extraction and reconciliation across contracts, invoices, bills of lading, and financial statements, with real-time discrepancy detection for customer and counterparty operations.

- Created and tuned context-aware prompt strategies on Gemini 1.5 Pro for high-precision extraction from unstructured trade and financial documents, mitigating hallucination risk on data where errors carry direct monetary cost.
- Designed reconciliation status checks combining LLM reasoning with sentence-embedding similarity matching to detect and report cross-source discrepancies in real time.

Technology Specialist

May 2021 - Jul 2024 | Pune, India

Built the core document-understanding model layer for an enterprise Document AI platform.

- Led the design, training, and evaluation of custom document-understanding models built on LayoutLM, Donut, and UDOP, achieving a 70% improvement in data extraction accuracy over the previous baseline.
- Spearheaded a Named Entity Recognition module, benchmarking multiple NER libraries and custom-trained models, and established the client-specific annotation and labeling workflows behind the training data.
- Implemented Donut-based DocVQA and LLM-driven approaches to extract from unseen and low-template document layouts without per-format rules.
- Owned model deployment pipelines end to end, containerizing models with Docker and shipping to AWS ECR/ECS with MLflow experiment tracking.

Engineering Lead

Aug 2018 - May 2021

Persistent Systems

Pune, India

Knowledge graph construction and conversational AI for enterprise data discovery.

- Led development of a knowledge graph linking organizations, individuals, and content across diverse data sources, modeled and served on Neo4j to enable relationship discovery.
- Built custom NER and relation-extraction classification models, and performed Named Entity Linking against Wikipedia using spaCy to populate and disambiguate graph entities.
- Developed the intent-classification engine for an automated customer-support chatbot using SVM, Naive Bayes, and RNN models combined with sentence-similarity features.

Systems Engineer to IT Analyst
Tata Consultancy Services

NOV 2013 - Aug 2018
Pune, India

Applied NLP/ML research and delivery for the ignio cognitive automation product and TCS Research.

- Built NLP and time-series pipelines that extracted structured, meaningful information from large-scale IT ticket data, computing key performance metrics and trends for the ignio team.
- Built a system that mines process and workflow models from system integration test cases, using NLP preprocessing and ML classification to recover software behavior from test artifacts.